



Listing multiples

Task A: Listing multiples

1) List the first five multiples of:

- a) 8
- b) 10
- c) 12
- d) 15

2) Alex says



30 is a multiple of 4 because
 $4 \times 7.5 = 30$

Is Alex correct? Explain your answer

3) Jacob started listing the first multiples of some numbers. He then spilled ink all over his work.

Find the multiples Jacob was working out and the missing numbers.

a) Multiples of 111 : 111, , , 444, 555, ...

b) Multiples of  : , 46, , 92, 115,  ...

c) Multiples of 324 : 324, , , , 1620, ...

d) Multiples of  : , , , , 4225, ...

Task B: Using listing to find common multiples

- 1) Find all the **common multiples** of 7 and 5 which are less than 100.

- 2) Find all the **common multiples** of 5, 8 and 10 which are less than 100.

- 3) Using the numbers, fill in the missing gaps. You can only use each number once.

5	6	9	10	12	12	27	27	50
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a) is a **common multiple** of and

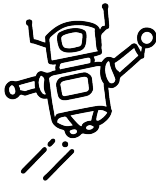
b) is a **common multiple** of and

c) is a **common multiple** of and

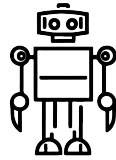


Task B: Using listing to find common multiples

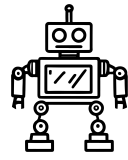
4) A path is made up from 100 steps, numbered 1 to 100.



Robot A
steps on
every 8th
step.



Robot B
steps on
every 3rd
step.



Robot C
steps on
every 6th
step.

a) Which numbered steps will all three robots have stepped on?

b) Robot D steps on every 2nd step. Will this change your answer to part a? Explain why.

Task C: Using common multiples to compare fractions

- 1) Using **common multiples** to compare fractions, identify the larger fraction in each pair below. You must show your working out.

a) $\frac{3}{5}$ and $\frac{2}{7}$

b) $\frac{4}{5}$ and $\frac{5}{6}$

c) $\frac{7}{10}$ and $\frac{3}{8}$

- 2) Three students have the following success rates on a game.

WELL DONE SAM! YOUR SUCCESS RATE IS $\frac{2}{3}$	WELL DONE LUCAS! YOUR SUCCESS RATE IS $\frac{11}{15}$	WELL DONE ALEX! YOUR SUCCESS RATE IS $\frac{3}{5}$
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Who is the best player? You must show your working out.

State a fractional success rate in between Sam's and Alex's success rates.

